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University students' sequential self-regulatory learning actions for completing cognitive-demanding and cognitive-less-demanding tasks with GenAI: ordered network analysis

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ABSTRACT

This study diverges from measuring university students' self-reported perceptions of Generative Artificial Intelligence (GenAI) by investigating their real-time self-regulatory learning (SRL) actions of interacting with GenAI. Self-regulatory learning actions refer to the process by which learners transform their mental abilities into using GenAI to achieve academic goals. Seventeen Chinese university students engaged in two tasks utilizing with the use of ChatGPT: one task involved translating a poem, characterized by a lower cognitive demand, and whereas the other entailed essay writing, which involved a higher cognitive demand. Students' computer screens were recorded, and a think-aloud protocol was implemented to capture both their cognitive processes and behaviours during the tasks. A total of 380 learning actions were recorded to understand how students self-regulated their learning with GenAI, and these actions were subsequently visualized using ordered network analysis. Although Zimmerman proposed three sequential SRL stages (forethought, performance and self-reflection), our findings indicated that university students did not sequentially follow these stages. These findings challenge existing teaching interventions that engage learners in thinking processes across sequential stages of SRL.

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1. Introduction

The continuous evolution of Generative Artificial Intelligence (GenAI) models highlights the transformative potential of artificial intelligence in education. Originally developed by OpenAI in 2019 (Thorp, 2023), GenAI, like ChatGPT, marked a significant advancement in natural language processing. Building on this foundation, the release of the ChatGPT-3.5 language model family in November 2022 introduced directive learning and reinforcement learning from human feedback, optimizing model performance (Abramski et al., 2023). This progress was furthered by the ChatGPT-4 family in March 2023, which brought enhancements in handling complex queries and customizing outputs (Skavronskaya et al., 2023). These advancements are believed to provide more individualized educational support for students, particularly in giving feedback, enabling more personalized and effective learning experiences (Stojanov, 2023).

GenAI has influenced university policies and practices (Cheng & Yim, 2024) and significantly transformed teaching and learning in higher education (Labadze et al., 2023; Rasul et al., 2023). While some scholars may worry that the use of GenAI could discourage students from developing problem-solving skills due to the ease of accessing information (e.g. Chan & Hu, 2023), we must recognize the value of the immediate feedback and abundant information provided by GenAI, as these features enable personalized educational paths tailored to the needs of learners (Stojanov, 2023). Therefore, the underlying challenge is how educators can guide students to use this tool to enhance their learning rather than becoming overly reliant on it (Lee et al., 2024), which leads to fundamental questions about how our students have adopted different learning approaches to use GenAI.

With the emergence of studies on the use of GenAI in higher education, many have focused on university students' perceptions of using GenAI (e.g. Chan & Hu, 2023; Strzelecki, 2024), including investigations into

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their motivation to learn with the tool (Hmoud et al., 2024) and online surveys about university students' acceptance and use of GenAI (Črček & Patekar, 2023; Wang & Gu, 2025). While these studies provide critical information on students' attitudes toward GenAI and their frequency of use, they fall short in objectively addressing how university students actually employ GenAI to complete various academic tasks. To our limited knowledge, a few research studies examine how students actually interact with GenAI. To fill this gap, the present study records the actual process of students using GenAI to complete various tasks, thereby providing a more objective analysis of their interactions with the technology, and examines how different self-regulated learning strategies were adopted.

1.1. Self-regulated learning and artificial intelligence

Self-regulated learning (SRL) refers to the process by which learners use their mental abilities to transform them into academic skills through self-directed goals (Zimmerman, 2002). SRL encompasses the skills, knowledge, and dispositions that students employ to manage and regulate various components of their learning (Hattie & Donoghue, 2016). These components include how individuals manage their cognition, behavior, motivation, and learning context (Kim et al., 2020), making SRL crucial for an individual to become a successful lifelong learner (e.g. Banihashem et al., 2025; Järvelä et al., 2016; Zimmerman, 2002).

Past studies confirmed that SRL is positively related to student learning outcomes in university (de Bruijn-Smolters et al., 2016; Tadesse et al., 2022), including online/blended learning (Luo & Zhou, 2024; Ma & She, 2024). In recent years, there has been a growing emphasis on researching SRL within online/blended higher education teaching and learning contexts (e.g. Cheng et al., 2025; Saadati et al., 2023; Urbina et al., 2021). While the effort of enhancing students' learning quality through technology is appreciated, more remains to be done in the field, especially when more advanced technologies, such as artificial intelligence (AI), within the SRL process is still largely underexplored (Lan & Zhou, 2025). Consequently, although research on SRL has been deeply rooted in traditional educational settings and has yielded substantial theoretical and practical insights, our study is situated within a technological perspective, specifically focusing on artificial intelligence.

AI has highlighted the potential of technological agents to interact autonomously with users to achieve specific goals, thereby transforming traditional approaches to learning and interaction. This development is shifting the paradigm at the intersection of education and technology, prompting an evaluation of how AI can be integrated into educational practices (Siemens et al., 2022). However, despite increasing interest in using AI to improve educational outcomes, there is a significant lack of research specifically focused on AI's role in SRL (Lan & Zhou, 2025). Among the limited existing studies, researchers emphasize the importance of further investigating the relationship between AI and SRL, recognizing that a deeper understanding could enhance the effectiveness of AI-supported educational tools. For example, Akinwalere and Ivanov (2022) highlighted the need to analyze methodologies and approaches for deploying AI effectively, ensuring that users benefit from these technologies while minimizing potential disadvantages. This raised concerns about the disconnection between theoretical frameworks of SRL and their practical application in AI-supported environments, warning that misalignment could lead to unreliable and inapplicable research outcomes (Kitto et al., 2024). This situation underscores the urgent need to better integrate SRL theory with practical AI applications to ensure the development of effective educational tools.

The urgency has escalated further with the breakthrough of generative AI (GenAI), renowned for its ability to provide immediate feedback from its extensive and versatile knowledge base, which emerges as a crucial educational resource to support students' SRL (Ng et al., 2024). Wu et al. (2024) examined enhanced self-regulation and knowledge construction among high school students using a GenAI-based Intelligent Learning Aid in a blended learning setting, compared to a control group. Similarly, Ng et al. (2024) provided evidence that GenAI can enhance students' knowledge, motivation, and SRL compared to a rule-based AI chatbot among high school students. These studies reinforce the notion that SRL is an important aspect of student learning that can be developed through technological support, particularly through tools like ChatGPT. Although the use of GenAI influences SRL, Karaoglan Yilmaz and Yilmaz (2025) found that SRLS has a notable positive impact on attitudes toward GenAI. This conclusion was based on a correlational study involving university students with GenAI experience, utilizing structural equation modeling analysis. While these recent studies offer valuable insights into the influence of SRL on GenAI, and vice versa, they

place a less emphasis on understanding how this impact occurs, leaving the underlying mechanisms as a black box.

Despite the limited number of studies focusing on SRL and GenAI within higher education, Wang et al. (2025) confirmed the effectiveness of GenAI feedback in fostering university students' self-reflection, goal setting, and strategy adjustment. Their study found significant improvement in SRL in the experimental group that utilized GenAI compared to the control group. However, these studies predominantly employed pre-post self-reported tests to illustrate SRL improvements, leaving the specifics of how students interact with GenAI largely unexplored. This exploration is as important, if not more so, than merely confirming the impact of GenAI on SRL. Dissecting how students engage with GenAI, which reflects their SRL, helps provide deeper insights into the mechanisms by which GenAI facilitates student learning.

AI-SRL research has expanded beyond simply focusing on and supporting SRL itself to also optimizing educational outcomes of learning activities (Banihashem et al., 2025). Wang et al. (2023) report that the cognitive load of learning tasks negatively affects performance, mediated by SRL in the process. According to cognitive load theory (CLT; Paas & Van Merriënboer, 1994), cognitive load is determined by the task's inherent complexity and delivery format. Considering that cognitive load interacts with SRL and influences learners' performance (Wang & Lajoie, 2023), which in turn affects how learning activities can be optimized, we also examine how individuals adopt SRL strategies when using GenAI to handle tasks with various cognitive loads.

1.2. Ordered network analysis to understand students' learning actions in GenAI

Ordered network analysis (ONA) is a novel analytic technique to examine students' sequential processes of learning, while only a few studies (e.g. Nguyen & Nguyen, 2025) used this technique to understand students' sequential learning behaviors. Through singular value decomposition and means rotation, ONA models a network of coded learning behavior, which can identify the frequency of connections between coded speech acts and temporal acts (Popov & Rochlen, 2025; Tan et al., 2024). To compute a model, ONA analyses the input coded data, visualizes the structure of temporal connections in a multidimensional space that allows researchers to statistically and visually compare two networks (Fan et al., 2023b; Tan et al., 2023, 2024). In the networks computed, ONA can visualize self-connection within a coded learning action through the thickness of nodes, as well as directional connections between two coded learning actions through the thickness of triangles (Tan et al., 2024).

By visualizing networks of temporal learning actions and comparing such learning actions between cognitive-demanding and cognitive-less-demanding tasks, ordered networks can provide insights into supporting university students' real-time interaction with GenAI. Regardless of the cognitive demands of the tasks using GenAI, such pedagogical support needs to facilitate students to critically reflect on and evaluate the GenAI outputs following an understanding of how GenAI is used (Ng et al., 2021; Stranges & MacNutt, 2025). Yet, more empirical evidence is needed to explore the strengths and weaknesses of students' sequential, real-time learning actions with GenAI, to design an instructional framework that supports students' lack of training in GenAI (Dai, 2025). The potential of ONA in shedding light on such strengths and weaknesses has been revealed in a study of a design program. Through self-connections revealed in the thickness of nodes, ONA revealed that students and professionals iteratively evaluated AI-generated ideas and reflected on designing moves to attain a higher quality of design (Nguyen & Nguyen, 2025). Strategically, ONA has been applied to compare two groups in terms of participants' design sketches, while the results showed that higher-scoring groups had a stronger sequential connection from evaluation and adaptation to reflection (Nguyen & Nguyen, 2025). Similarly, the application of ONA in comparing students' use of GenAI to complete cognitive-demanding and cognitive-less-demanding tasks. This can be revealed through self-connections of a real-time learning action through the thickness of the nodes, as well as sequential connections between two learning actions through the thickness of the chevron.

1.4. Current study

The development of GenAI aligns with the growing demand for accelerated integration of technology into online and blended education environments (Jin et al., 2023; Lee et al., 2024). Guided by Zimmerman's SRL

model, which was the most frequently applied theory among AI-SRL studies (Banihashem et al., 2025), our study utilizes the SRL model to categorize and analyze student interactions with GenAI during various academic tasks, focusing on their engagement behaviors. The SRL model encompasses three critical phases: forethought, performance, and self-reflection (Zimmerman, 1990). The forethought phase is the initial stage where learners analyze the learning task, set specific goals, and devise strategies to accomplish them, laying the groundwork for the entire learning process. The performance phase involves learners actively engaging in learning activities while continuously monitoring and adjusting their behaviors to ensure alignment with their learning objectives. Finally, the self-reflection phase occurs when learners review and evaluate their learning outcomes, assessing their effectiveness and identifying areas for improvement (Pintrich, 2000; Zimmerman, 2008). These structured phases provide a comprehensive framework for examining students' approaches to completing different learning tasks.

This research aims to provide educators and instructional designers with insights into how university students adopt different SRL approaches when using GenAI for academic tasks that require varying levels of cognitive demand. "Learners adjust what they do and how they study depending upon task, self, and context conditions" (McCardle & Hadwin, 2015, p. 45), while keeping the individual and the study context as a control, this study explores how students adopt different SRL when they encounter different tasks. The findings will inform decisions regarding task difficulty and the development of strategies that enhance student interaction with AI tools like ChatGPT, including the enhancement of students' SRL. Below are the research questions:

RQ1. Are there any differences in the number of learning actions when university students interact with GenAI to complete the poem-translation and essay-writing tasks?

RQ2. As indicated by ONA, what are the similarities and differences in the order networks of learning actions of university students between completing the poem-translation task and the essay-writing task?

2. Methodology

To facilitate an in-depth exploration of participants' cognitive processes and interactions with ChatGPT, we employed a task-based approach, utilizing the think-aloud protocol and screen recording as primary data collection methods. The task-based approach was chosen because it provides a structured yet naturalistic setting, enabling us to directly observe how participants interact with ChatGPT while completing specific tasks (e.g. Lytovchenko et al., 2025). Screen recording was implemented to capture both participants' reactions and their entire work screen, ensuring no data was lost during the process. Additionally, the think-aloud protocol was adopted to enrich the data collected. This method is widely used in cognitive and usability research, involves participants verbalizing their thought processes while performing tasks, and offering valuable insights into their cognitive strategies and decision-making (e.g. Angelone, 2025). Ethical approval for this study was obtained from a public university in Hong Kong.

2.1. Participants

We recruited 17 volunteers (comprising 8 males and 9 females) who were Chinese students enrolled in master's programs at a public university in Hong Kong during the 2023–2024 academic year (Table 1). To avoid any role conflict, we did not recruit the participants ourselves, as we were mindful of our academic positions at the university and wanted to prevent any undue influence or power dynamics with the students. Recruitment was conducted using convenience and snowball sampling methods; we invited three student helpers to recruit other participants based on their personal networks at the university.

Participants were eligible for the study if they were master's students whose primary language was Chinese and were enrolled in a master's program at the university. Compared with undergraduate students, master's students are presumed to possess a greater ability to articulate their thoughts, making them more suitable for completing the two learning tasks while adopting a think-aloud approach in this study. This particular group was selected not only because the majority of non-local students in Hong Kong are native Chinese speakers from Mainland China but it also helps control several factors, including (i) they reported that they did not use ChatGPT as ChatGPT is inaccessible in China, and (ii) the students' English proficiency,

Table 1. Participants' demographics.

No	Sex	Enrolled discipline	Required time to complete the two tasks
S1	Female	Social Sciences	13.55 min
S2	Female	Housing	3.07 min
S3	Male	Social Sciences	5 min
S4	Female	Applied Psychology	8 min
S5	Male	Science	4.44 min
S6	Male	Computer Engineering	9.36 min
S7	Male	Data Science	9.35 min
S8	Female	Economics	13.12 min
S9	Male	Social Sciences	21.4 min
S10	Female	Social Sciences	27.39 min
S11	Male	Science	31 min
S12	Male	Social Sciences	27.56 min
S13	Female	Language	11.26 min
S14	Female	Social Sciences	15.58 min
S15	Female	Social Sciences	17.14 min
S16	Male	Computer Engineering	37.56 min
S17	Female	Social Sciences	22 min

since the learning task involves translating English. To enroll in this university, students are required to have a minimum IELTS Academic band score of 6.5 (with an average between 6.5 and 7), ensuring their English abilities are relatively consistent.

2.2. Research design

The first author trained the three student helpers to conduct the research according to a protocol to ensure procedural consistency. Participants were required to complete two tasks using ChatGPT online. Interested and eligible individuals would receive a Zoom or Tencent meeting link to join the study. Upon entering the online meeting room, they were briefed on the study and asked to provide consent. Once consent was obtained, the student helpers introduced the participants to the think-aloud protocol. After participants demonstrated their understanding of the think-aloud protocol, they received a Word file containing two tasks (details of the tasks are in section 2.3), which they were instructed to complete. To encourage a more self-directed approach, student helpers reminded participants during the briefing that they were simulating real university assignments and that their goal was to complete the tasks using AI and attain high marks.

Participants could decide which of the two tasks to do first. After completing one task, student helpers would double-check with them whether they had finished and remind them of what to do for the remaining task, thereby providing a washout period to minimize potential carryover learning effects.

The screen-sharing function was utilized to record their entire screen during task completion. Participants were given access to the Word file and ChatGPT, but they were also allowed to open other screens they deemed helpful for completing the tasks. The entire process was documented, with videos averaging 19.9 min in length. Across the 17 participants, a total of 380 turns were captured.

2.3. Tasks design

To design tasks with one high and one low cognitive load, we employed Blooms taxonomy (1956) to decide on the task complexity. In this taxonomy, "knowledge" is considered the most basic level of cognitive learning. However, this level was intentionally not chosen because it was deemed too simple for university post-graduate students, who might not even need to use ChatGPT to complete such tasks. To create tasks that mimic a typical university context, the widespread use of ChatGPT among Chinese students attending English-medium universities was considered. These students often need to submit assignments in English, making it common for them to use ChatGPT for translation, which falls under the second level of Bloom's taxonomy, comprehension. Additionally, it was considered that essay-based assignments are the most common form of university work and align with the second highest level in Bloom's taxonomy in terms of higher-order cognitive skills, synthesis. Considering these factors, two types of assignments were selected: poem translation (Task 1), which is considered less cognitively demanding, and essay writing

(Task 2), which is more cognitively demanding. There are no restrictions on the order in which the tasks are completed; students may choose which to do first.

According to CLT, prior knowledge can also affect the cognitive load of a task (Paas & Van Merriënboer, 1994). To account for the participants' diverse academic backgrounds, the two tasks are designed to be non-subject-specific, minimizing content-related variations in cognitive load. Task one involves translating the English poem "The Road Not Taken", which is unfamiliar to many Chinese participants; during the task, it was confirmed that all participants had not heard of this poem before. As a result, they must understand the words when using ChatGPT for translation. Task two requires participants to write a short essay offering advice to graduate students seeking employment in Hong Kong, with no word limit. Since most participants are non-locals unfamiliar with the local job market, they will rely on ChatGPT to gather relevant information and critically evaluate the validity of its suggestions while composing the essay. Although ChatGPT can provide answers for both tasks, participants must actively process and interact with the responses, adopting different SRL strategies to handle the ChatGPT-generated suggestions effectively.

2.4. Coding process

Building upon the methodology established by Fan et al. (2023a), which triangulates and integrates the measurement of self-regulated learning (SRL) using trace and think-aloud data, this study draws upon their framework for developing an action library to code learning actions (see Table 2). Given that our research focuses on interactions with ChatGPT, the actions identified are more specific, such as "Copy and Paste (To GenAI)" and "Copy and Paste (Within GenAI)". We captured each action carried out by the participants on the computer while completing the two tasks with ChatGPT. All the actions are labeled according to their actual sequence.

After defining each action label with its corresponding description, we categorize these labels into the three critical phases of the SRL's cyclical phases model (Zimmerman, 1990): forethought, performance, and self-reflection. Since this study focuses on how student learners interact with ChatGPT for learning purposes, adopting the SRL model provides a systematic way to analyze and understand these interactions. The forethought category includes learners' actions that orient and plan to accomplish tasks, such as (re)reading the task instructions and navigating through pages to review the tasks. The performance category encompasses learners' actions that involve actively engaging in the learning activities, such as giving and revising instructions to ChatGPT. The self-reflection category pertains to learners' actions that review and evaluate the content generated by ChatGPT, such as using other search engines or software to review and cross-check information.

The recorded videos were transcribed by two student helpers who were not involved in the data collection process, thereby providing unbiased perspectives on the data. Each student helper independently processed all the videos, beginning with a verbatim transcription of participants' utterances during task completion. Following transcription, they applied codes from the action library (Table 2) to classify the learning actions. To verify the comprehensiveness of the action library, the first author instructed the assistants to initially code five videos. Once the action coding was refined, the student helpers completed the remaining transcription and coding independently. Subsequently, these actions were then organized sequentially and paired with participants' utterance as demonstrated in Fan et al. (2023a)'s study, to provide a full picture for

Table 2. Coding Library for labeling learning actions.

SRL categories	Action labels	Action descriptions
Forethought	Copy and Paste (To essay)	Learners copy and paste content from Gen-AI to the essay
	Copy and Paste (To GenAI)	Learners copy and paste content from the essay to Gen-AI
	Copy and Paste (Within GenAI)	Learners copy and paste content from Gen-AI to Gen-AI
	Task Instruction	Learners read or re-read the given task.
	Navigation	Learners navigate through pages or scroll.
Performance	Give Instruction	Learners are giving instructions to Gen-AI for the first time for a task.
	Revise Instruction	Learners revise the instructions to Gen-AI after getting a response.
	Write Essay	Learners write, edit or stay on the essay page.
Self-reflection	Review Result	Learners review the generated results.
	Search Content	Learners use a search engine to search for content.
	Cross Check	Learners use other software to cross-check the generated content.

us to decide which phase of the SRL model, forethought, performance, or self-reflection, they belonged to. The resulting sets of codes and transcriptions were cross-checked, demonstrating high intra-rater reliability (96%), and any discrepancies were addressed through discussion until consensus was reached.

2.5. Data analysis

To answer RQ1, we computed the frequency of learning actions between the two tasks: the poem-translation tasks and the essay-writing tasks. We sorted the eleven learning actions into three different increasing levels of cognitive demand according to the conceptualization of three phases of self-regulatory actions: forethought, performance and self-reflection. Chi-square tests were used to statistically compare the frequency of learning actions under these three components of GenAI literacy was significantly different across the two tasks.

Also, to answer RQ2, the R ONA package (Tan et al., 2024) was used to model the sequential connection among eleven learning actions (Table 2). For each turn, we identified their presence and absence in an Excel spreadsheet. After coding the responses, ONA identified temporal connections between two consecutive learning actions (for details, please refer to Fan et al., 2023b). These sequential connections between learning actions formed a network by ONA (Fan et al., 2023b). A bidirectional connection, which is indicated by a pair of triangles, indicates the connections between two different nodes. A chevron will only appear in the triangle with a stronger directional connection within the paired bidirectional connections. Three ordered networks were created: one network characterized sequential learning actions in the poem-translation task; the other network characterized the sequential learning actions in the essay-writing task; the final network is a subtracted network that characterizes the differences in sequential learning actions between the poem-translation and essay-writing tasks. To statistically compare two ordered networks between the poem-translation and essay-writing tasks, we run a t-test to compare two networks (Tan et al., 2023).

3. Findings

3.1. Number of learning actions (RQ1)

As shown in Table 3, there was a statistically significant difference in learning actions classified into the phase of forethought learning action ($\chi^2 (3) = 13.4, p < 0.01^{**}$). In the poem-translation task, there was a higher frequency of copying and pasting GenAI outputs to the essay ($n = 8, 4.0\%$), as well as copying and pasting text within GenAI ($n = 5, 2.5\%$). By contrast, in the essay-writing task, there was a higher frequency of copying and pasting texts to GenAI ($n = 14, 8\%$) as well as navigating the GenAI interface ($n = 4, 2\%$). It might be because students understood that simply copying and pasting texts to other sources was not acceptable for essay-writing tasks, but it was acceptable for the translation task. Students also spent more time navigating GenAI outputs, as essay-writing requires more thinking compared to translating a poem.

Another observation was that there was no significant difference in learning actions between the two tasks in the phase of performance ($\chi^2 (1) = 0.461, p = 0.794$) as well as self-reflection ($\chi^2 (3) = 0.526, p = .769$).

Table 3. Distribution of GenAI learning actions across tasks.

Self-regulatory model	Learning actions	Number (Percentage)		Chi-square test results
		Translation (Task 1)	Essay-writing (Task 2)	
Forethought	Copy and paste (To essay)	8 (4.0%)	3 (2%)	$\chi^2 (3) = 13.4, p < 0.01^{**}$
	Copy and paste (To GenAI)	3 (1.5%)	14 (8%)	
	Copy and paste (Within GenAI)	5 (2.5%)	1 (1%)	
	Navigation	1 (0.5%)	4 (2%)	
	Task Instruction	16 (7.9%)	19 (11%)	
Performance	Give Instruction	17 (8.4%)	14 (8%)	$\chi^2 (1) = 0.461, p = 0.794$
	Revise Instruction	33 (16.3%)	20 (11%)	
	Write Essay	21 (10.4%)	15 (8%)	
Self-reflection	Review Result	75 (37.1%)	68 (38%)	$\chi^2 (3) = 0.526, p = .769$
	Search Content	9 (4.5%)	10 (6%)	
	Cross Check	14 (6.9%)	10 (6%)	
Total		202 (100%)	178 (100%)	

$p = 0.769$). This might be because the literacy at higher cognitive levels involved in both tasks was similar for both tasks. For example, in the phase “Self-reflection”, students needed to search if the translation by GenAI was correct (Task 1), as well as cross-check if the suggested points in the essay were valid (Task 2).

3.2. Ordered network in two tasks (RQ2)

3.2.1. Statistics in ordered networks

The coded learning actions of the two tasks were projected into multidimensional spaces by ONA, with each task indicated by the loci of the arithmetic mean (indicated by the square with error bars in Figure 1). For the translation task, the locus of the projected points of the poem-translation task was $(0.05, 2.34 \times 10^{-17})$, which was positioned at the positive side of the vertical and horizontal axes. On the other hand, the locus of the project of the essay-writing tasks was $(-0.06, -1.15 \times 10^{-17})$, which was positioned at the negative coordinate of the vertical and horizontal axes.

A Welch Two-Sample t-test was conducted to identify if there were any statistically significant differences between the ordered networks of two tasks along vertical and horizontal axes. Along the horizontal axis, there was a statistically significant difference ($t(379.12) = 6.3035, p < 0.001$, lowest 95% CI = 0.072, highest 95% CI = 0.137). However, along the vertical axis, there was no statistically significant difference ($t(391.82) = 1.04 \times 10^{-15}, p > 0.05$, lowest 95% CI = 2.33×10^{-17} , highest 95% CI = -1.15×10^{-17}). In other words, the projected coded learning actions have significant differences along the horizontal instead of the vertical axis.

3.2.2. Ordered network of the poem-translation task

To meaningfully analyze such differences by interpreting the distribution of coded learning actions, we plotted an ordered network of coded learning actions of the poem-translation task, that of the essay-writing task, as well as a subtracted network by aligning both tasks (Figure 2). The subtracted network indicates the differences in sequential connections between the tasks involving students’ interaction with ChatGPT. For the poem-translation task, students mostly focused on the task instruction and then reviewed result. This was indicated by the chevron pointing from the node “Search Content” to “Review Result”. Such a finding indicated that after students used a search engine to search for content to verify GenAI outputs, they critically reviewed GenAI outputs. Thus, the two learning actions manifesting the same phase of self-reflection were sequentially connected.

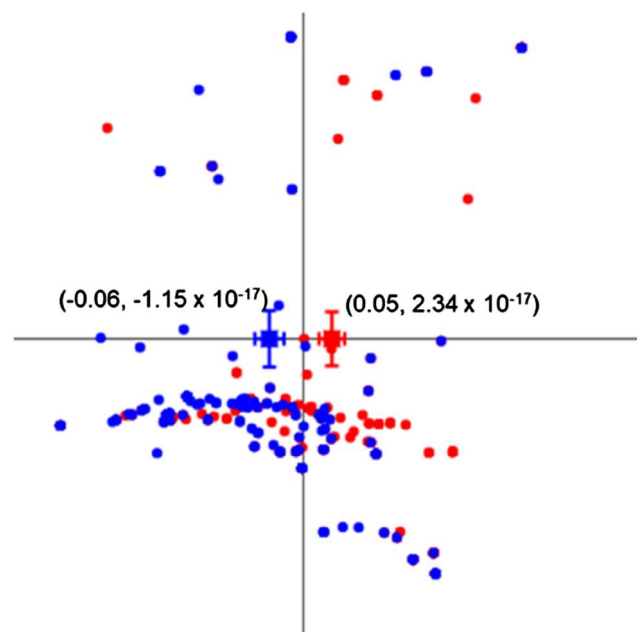


Figure 1. ONA points (dots) and their mean point (square) for the poem-translation task (red) and the essay-writing task.

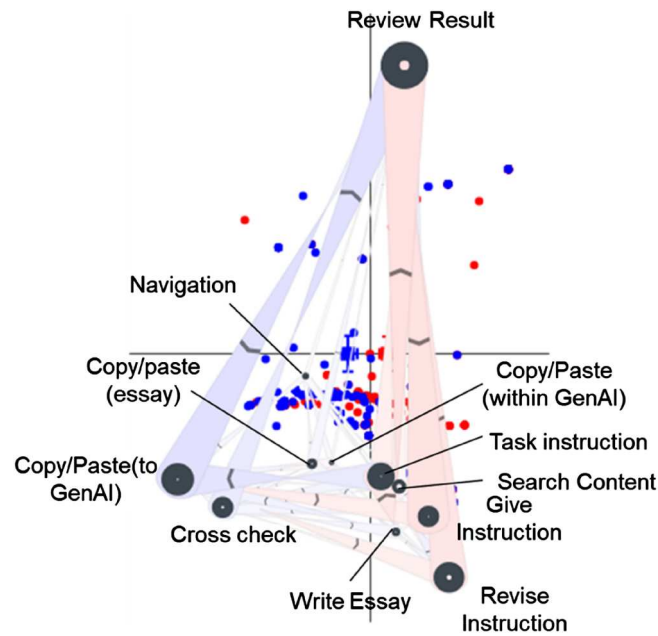


Figure 2. ONA subtracted networks between the poem-translation task (red) and the essay-writing task using ChatGPT (blue).

The consecutive connection from “Search Content” to “Review Result” is exemplified in Table 4. In Turn 5, student 03 was unsure about the translation version by ChatGPT. The student searched Baidu and 360 to verify if the translation by ChatGPT was correct. Following turn 5, students compared the translation by ChatGPT and the search engine and verified the translation by ChatGPT. As seen in the transcript, the student critically evaluated GenAI outputs in different ways, such as searching for content and reviewing GenAI outputs.

Also, apart from shifting between learning actions of the same phase of self-regulatory learning action, students reviewed outputs of ChatGPT regarding poem translation, then either gave instructions or revising instructions of ChatGPT. This was also justified by the chevron connected from “Review Result” to “Give Instruction”, as well as “Review Result” to “Revise Instruction”. “Review Result” is a component of self-reflection, while “Give Instruction” and “Revise Instruction” are components of performance. Following self-reflection, students then perform their academic skills in GenAI in real-time. This showed that students’ interaction with GenAI in translating the poem can dynamically shift across different phases of self-regulatory learning.

An example excerpt is indicated in Table 5. In the excerpt, Student 17 engaged with a sequential learning action from reviewing result to revising the instruction. For example, in Turn 5, the student reviewed the results of ChatGPT which did not give basic information about the poem, and the student suggested that a synergistic use of different GenAI apps would be useful for the task. Following that, the student revised the instruction to GenAI by typing “I want to make it smoother and modify it, please” (Turn 6), and critically reviewed the result that ChatGPT didn’t help the student to correct the passage (Turn 7). Following a critical evaluation of GenAI output, student 17 continued to revise the instruction by entering “please modify it, and there are two requirements, the one is making it smooth, the other is making it looked less translation accent”. This showed that the students manifested self-regulatory learning actions shifted between the phase of performance and the phase of self-regulation.

Table 4. Episode of Student 03’s interaction with GenAI in the poem-translation task.

Turn	Coded actions	Transcript
5	Search Content	Can try to search it on 360 or Baidu. We see how they translate their point.
6	Review Result	The road not taken. There are both Chinese and English versions. I see it again. When I open it again, they have “The Road Not Taken”. Yes, it is correct. It pointed out the characters in the poem. It also points out that in the yellow forest, there are two separate roads.

Table 5. Episode of Student 17's interaction with GenAI in the poem-translation task.

Turn	Coded learning actions	Description of actions	Transcript
5	Review Result		Sometimes GPT gets stuck. I often try different software. Sometimes I will try an AI software called "Tinggong" produced by my country. Sometimes, if one AI software cannot be used, I will combine another one to use it. "Tinggong" will first give the basic information about this poem. Okay.
6	Revise Instruction	Enter "I Want to make it smoother, modify it please"	Okay. The first instruction lets GenAI help me revise a coherent passage.
7	Review Result		This AI software doesn't understand my meaning. It only gives me some general responses. I want him (GenAI) to revise this passage.
8	Revise Instruction	Enter "please modify it, and there are two requirements, the one is making it smooth, the other is making it looked less translation-accent"	Let me instruct it again.

3.2.3. Ordered network of the essay-writing task

Contrasting with the poem-translation task, students' learning actions with ChatGPT in the essay-writing task shifted between the self-regulatory phase of forethought and self-reflection. In Figure 2, there was a chevron in a directional triangle pointing from "Review Result" to "Copy/Paste (to GenAI)". This showed that students initially demonstrated their GenAI literacy by self-evaluating GenAI outputs, followed by forethought with GenAI. Their manifested learning actions were said to revert from a higher cognitive level to a lower cognitive level of GenAI literacy. Moreover, there was a chevron in a directional triangle pointing from "Cross check" to "Review Result". This justified that university students stayed at the level of self-evaluating GenAI, instead of putting into action in performing with GenAI.

The temporal connection was further illustrated in the excerpt in Table 6. Student 11 copied the task question to ChatGPT (turn 3) and then reviewed the problematic results in GPT (turn 4). However, the student did not reflect on the reasons for such problems and repeated the learning action of copy and paste to GenAI. Although the outputs seemed to be problematic, the student did not revise the instruction, while not performing with GenAI, regardless of the students' critical evaluation of GenAI outputs. This showed that the student did not momentarily adjust their self-regulatory learning actions.

Similarly, student 11 was engaged with repetitive cycles of learning actions of reviewing results and followed by cross-checking (Table 7). In other words, the students were able to self-evaluate GenAI outputs, but they did not perform on GenAI. In turn 24, the student critically examined the output by Bard and deemed that the learning and career advice provided by Bard was holistic. Following reviewing result, student 016 also cross-checked the learning and career advice with different GenAI tools, such as ChatGPT, Bard and Ernie Bot (turn 25). From turns 26 and 27, student 11 positively appraised the semantic meaning by different GenAI bots, followed by cross-checking the use of terminologies across different bots. Yet, the

Table 6. Student 11's interaction with GenAI in the essay-writing task.

Turn	Coded learning actions	Description of actions	Transcript
1	Task Instruction		(I will choose the second question. Will choose the second question. I came to Hong Kong to study for graduate school, and I am not very adapt ...) Please give me some specific suggestions on how to deal with conflicts with roommates and group members after giving specific suggestions. Please generate relevant answers multiple times based on your major. This has to be based on our major.
2	Copy Paste (to GenAI)	copy question 2.1" I came ... group member"	I will raise questions to GPT. And then copy to there, how to deal with companies, roommates, and group. This is the question I give GPT. I have to tell GPT my major. My major is to automate my major specialism.
3	Give Instruction	input" I major is intelligence automatic"	Yes. Give the question to GPT, and see how GPT responds.
4	Review Result		GPT seems to have some problems. I have already encountered this problem. Refreshing it would be great. Refreshing one more time is then good.
5	Copy Paste (to GenAI)	copy question 2.1" I came ... group member"	Then we copy and paste the same question. Similarly, we have to patiently wait for ChatGPT. Now it said, giving 9 points of opinion, and then 10 points of opinion. We have to select these questions, and give the same questions to Ernie Bot, a similar bot to Bard.

Table 7. Student 11's interaction with GenAI in the essay-writing task.

Turn	Coded learning actions	Description of actions	Transcript
24	Review Result	bard result	Then the following are specific issues related to your failure. This means Bard has noticed my major, my specialty, as well as my major study, habits and schedules. It also specifies some learning plans and habits, as well as plans for collaboration and projects. These things are needed as it considered my expertise. Bard also provided some suggestions according to my learning. Then it also offered additional tips, such as building connections and communicating with other students. Seek, support.
25	Cross Check	Comparing Ernie bot, ChatGPT and bard	Three of them mentioned this. Their ways of saying this are different, but the meaning is the same. There is one point which is very important. Then focus on self-care. It is the same. Then they said compromise. But they, at the same time, mentioned "focus on self-care". At these two places, they mentioned "compromise". Some people might not like to compromise because they have their own principles and their own characters.
26	Review Result	bard result	At this place, it is mentioned that we need to respect our own thinking. In dealing with some small problems, we must compromise, solve the problems, and integrate into the social community. But for some basic problems, we have to insist on our own principles. (Eventually, we can see how Bard concludes.) Remember, adapting to a new environment takes time, and effort, be patient with yourself and others and focus on finding some solutions that work for everyone.
27	Cross Check	Compare Bard, Ernie bot and GPT	The conclusion here is not different from that of Ernie Bot. Apart from carefully using some words, the meanings are similar. When we see the conclusion by GPT, we can read it again. Remember that into a new environment takes time, be kind to yourself and give yourself and others time to adjust to your experience in Hong Kong for your graduates. Studies can be enriching and rewarding with the right approach. The conclusions offered by three chatbots are similar. (There is some encouragement and shortening of the conclusion in the first part. What they said was very positive, making some wishes to us. Just like this sentence, I hope these suggestions are helpful. Good luck with your studies in Hong Kong.
28	Write Essay	input : 1. (If we really help, we seek help). 2.	So in conclusion, this question gives me some answers. I will summarize the opinions of these three helpers. Here, there is no title. This title I won't edit its name. The answer is the important that if we need to seek help, seek help. If I need to seek help, seek help. I understand the second point. Regarding the third point, I get it. I think the suggestion is suggestive in nature. This suggestion is a guiding light, a light tower. They did very well on the framework, the blueprint. That means I think their outputs are good. I will think about the details myself.

student gave a very simple suggestion for study advice in turn 28. From this episode, the student did not demonstrate much critical self-reflection on the GenAI output, as the student was satisfied with the GenAI outputs from three bots. The suggestion written down in turn 28 was simple. Hence, the students did not revise any prompt instruction. The episode stayed at the level of self-reflection, while the student did not transition to performing with GenAI.

3.2.4. Summary of the difference in ordered networks between the two tasks

The findings from the ordered networks were summarized in the two tasks (Figure 3). In the poem-translation task, there were learning actions that were sequentially connected within the stage of self-reflection, as well as across the stages of self-reflection and the level of performance. This was important for students' informed use of GenAI as students were able to self-evaluate the outputs of GenAI, then give and revise instructions in the GenAI. In the essay-writing task, it was not common for students to perform with GenAI; rather, they understood basic forethought learning actions of GenAI, such as reading task instructions and copying and pasting texts to GenAI. Even though they demonstrated their self-reflection of GenAI outputs, it was not common for them to give and revise the instruction.

4. Discussion

This study characterized students' in-the-moment learning actions with the phases of the SRL model by coupling think-aloud interviews and screenshotted video-recordings. Innovatively, this study used ordered network analysis to characterize the sequential connections between learning actions in a cognitively demanding task and a cognitively-less-demanding task.

Despite the increasing focus on how to utilize GenAI to enhance students' SRL (Self-Regulated Learning) capabilities, most studies have primarily discussed the connection between GenAI and SRL rather than exploring the specific mechanisms or processes involved (Sardi et al., 2025). This study is among the few

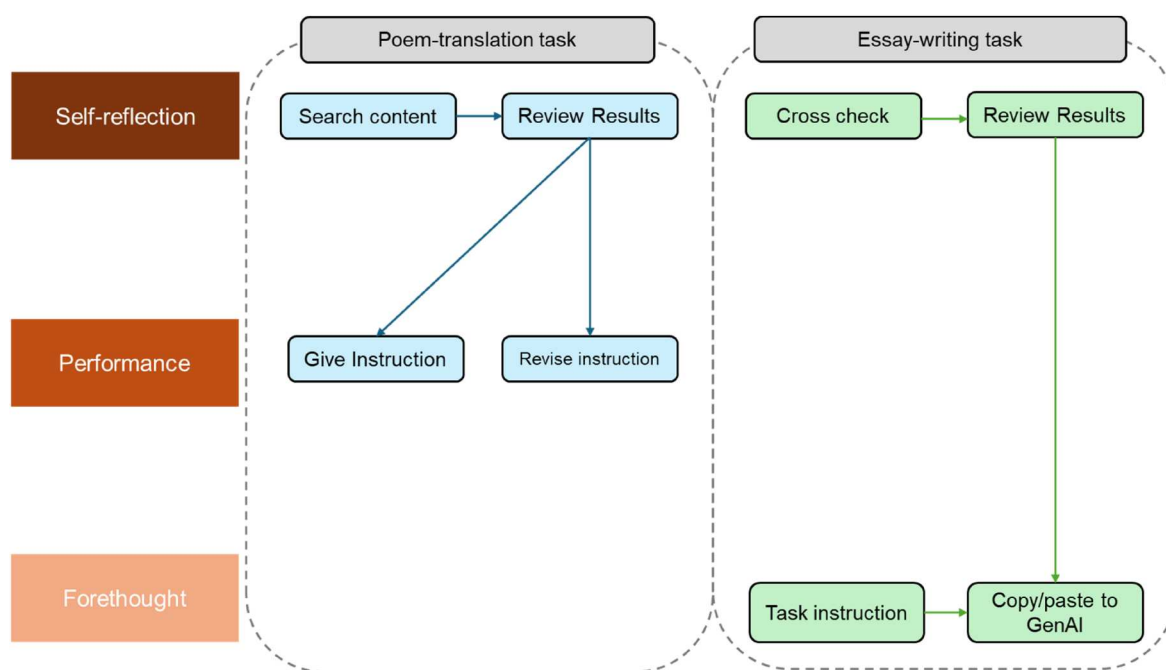


Figure 3. A summary figure of the two tasks (poem-translation and essay-writing tasks).

that dissect how students interact with ChatGPT across different learning tasks, allowing us to better understand the processes underpinning SRL. While Zimmermans cyclical phases of learning identify three sequential stages: forethought, performance, and self-reflection (1999), our findings suggest that the use of ChatGPT may not strictly follow this progression. Specifically, self-reflection appears to lead students toward either performance or forethought. This indicates that, with GenAI tools like ChatGPT, students might spend less effort on understanding or planning their tasks upfront. Instead, they may first use ChatGPT to generate a preliminary result, which then informs their subsequent planning (forethought) or immediate action (performance).

We also observed differences in the adoption of SRL depending on the varying cognitive loads of the tasks. The essay writing task, presumed to be cognitively more demanding based on Bloom's taxonomy, did not result in a greater number of learning actions compared to the poem-translation task. The findings indicate that participants demonstrated lower performance using and applying GenAI (performance) in the essay writing task, such as given instructions or revise instruction. This behavior suggests that learners predominantly relied on the initially generated outcomes provided by ChatGPT without actively prompting the AI for refinements or enhancements. In contrast, the poem-translation task elicited more dynamic interaction across different phases of self-regulatory learning, namely self-reflection and performance. Participants offered specific prompts, such as "Make the last word of the first and last sentence of each paragraph rhyme" (Student 10) and "translate more elegantly" (Student 15) (performance). Additionally, some students took the initiative to utilize other translation tools, such as DeepL and Baidu translation, to cross-verify the outputs (Students 1, 4, 10, 11, 13, 16) (self-reflection). Although Wang et al. (2023) investigated the link between cognitive load, student performance, and the role of SRL, our research does not focus on comparing performance outcomes. Instead, it seeks to examine variations in participants' approaches to tasks. Additionally, our study adds to the sparse body of research on GenAI by exploring AI's influence on SRL (Lan & Zhou, 2025), particularly with tasks with different cognitive loads.

The varied cognitive demands of the two tasks have led to very different generated outputs from GenAI, which in turn affected students' SRL approaches. In response to the essay-writing task, GenAI yielded multiple points as a response and provided detailed explanations. It is observed that participants tend to directly choose several points to adopt from the generated results without prompting for more answers from GenAI, perceiving the initial generated answers as relatively comprehensive and exhaustive. This behavior suggests a reduced inclination to critically evaluate or seek alternative solutions, which aligns with concerns regarding

the potential negative impact of GenAI on higher-order thinking skills (Lee et al., 2024). Participants are more inclined to accept GenAI's initial responses when faced with a cognitively demanding task. In contrast, when the task is less cognitively demanding and the generated outputs are short, the participants tend not to consider the initial response from GenAI as a perfect or final answer. This indicates that employing GenAI for tasks with lower cognitive demands may be more advantageous for student SRL compared to tasks with higher cognitive demands. Aligned with the AI-SRL research focus on optimizing educational outcomes of learning activities (Banihashem et al., 2025), this study demonstrated potential differences in students' SRL when using ChatGPT across various tasks. It emphasizes that SRL with ChatGPT should not be viewed as a uniform action and underscores the importance of further categorizing different types of tasks based on various dimensions of cognitive load (Wang & Lajoie, 2023) to better examine their potential differences.

Methodologically, ordered network analysis can visualize temporal connections between learning actions, which leads to the generation of a conceptual model that characterizes students' learning with GenAI, address existing research gap by linking the SRL theory and AI practice better (Akinwalere & Ivanov, 2022; Kitto et al., 2023). For example, ONA shows if learning actions move dynamically across phases of self-regulatory learning. Interestingly, the essay-writing tasks have a different dynamic trajectory in self-regulatory learning actions compared to the poem-translation task, with the self-regulatory learning focus on self-reflection, then forethought. Whereas in the poem-translation task, students perform self-reflection first and then performance. Not only can ONA be applied to different ability groups in navigating GenAI (Nguyen & Nguyen, 2025), but it also compares students' real-time learning actions across tasks with different cognitive demands. In the future, ONA can be potentially applied to comparing real-time learning actions of tasks of different disciplines.

4.1. Limitations

Our study has several limitations that should be considered when interpreting the findings. First, the sample size was relatively small and limited to a single public university in Hong Kong. However, the sample was intentionally not restricted by subject, gender, or study year to enhance the generalizability of the results to a broader population of Chinese university students. Additionally, this study is cross-sectional, and the data were collected during a period when only ChatGPT 3.5 was freely accessible to students through the university. However, since participants were not restricted to using only the university-provided ChatGPT, some participants, such as S18, may have accessed ChatGPT 4 independently. With the emergence of newer developments, future research could also consider the impact of different versions and access levels to ChatGPT. Hence, the data were collected in the first quarter of 2024, a time when ChatGPT was still relatively new to many users. Future studies could explore whether SRL approaches with ChatGPT evolve as learners become more sophisticated and experienced with the tool. Furthermore, this study only adopted one set of data to capture the essence of students' SRL behavior in using ChatGPT; future studies can collect different datasets or multiple sources of data to obtain a more comprehensive understanding. Finally, although this study considered participants' gender and previous experience with ChatGPT, it did not account for perceptual factors that might influence how participants utilize GenAI, such as their attitudes toward SRL. Mirriahi et al. (2025) found that SRL constructs of self-efficacy and social support impact perceptions of ease of use and actual engagement with GenAI. Consequently, future research could further investigate how these perceptual variables affect students' use of GenAI.

4.2. Implications for research and practice

The findings of this study have implications for research, policy and practices. Regarding contribution to the field, this study highlights several key implications for educators in leveraging GenAI as an educational resource to enhance students' SRL. The sequential differences in SRL approaches adopted when using GenAI for tasks of varying cognitive levels underscore the importance of considering the complexity of the tasks assigned. Rather than viewing the impact of GenAI on SRL as uniform, researchers should examine the specific cognitive demands of the tasks involved during their study of university students' interaction with GenAI. Many existing studies have explored how to facilitate through GenAI in the context of personalized learning approaches (e.g. Lai, 2024; Xia, 2025). Building upon this body of research, our

study extends the investigation from individual learners to learning groups, taking into account the varying cognitive loads associated with different tasks. This broader perspective can help inform how SRL can be effectively fostered within collaborative learning environments.

The findings also have implications for practice and policy. When educators teach students how to use GenAI to facilitate their essay writing, university teachers need to devote more time to teaching students how to give and revise GenAI prompts. This is supported by research findings that students' performances according to the self-regulatory model are less likely to be observed in the essay-writing task. Also, at a policy level, our findings suggest that university assignments and coursework should require students to report how they give and revise GenAI prompts during GenAI-assisted essay writing. University students simply review the GenAI output and copy-and-paste as their own work without demonstrating critical thinking. University assessment policies should be clear in terms of the percentage of work and the minimum number of GenAI prompts that university students need to give before handing in their essay-writing tasks. This prevents university students from simply reverting from self-reflection to forethoughts, without demonstrating the "performance" level according to the self-regulation model.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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